A Small Tower of Near-Ramanujan Biregular Graphs

Abstract

For every prime power q we give an explicit infinite family, indexed by $s = 2, 3, 4, \ldots$, of (q, q^3)-biregular bipartite graphs. The normalized second singular value is at most $2/\sqrt{q}$, uniformly in s. Thus, if $\Delta = q^3$ denotes the larger degree, the unnormalized second singular value is at most $2\sqrt{\Delta}$. The parameter s is essential: for fixed q the degrees stay fixed while the number of vertices tends to infinity.

1 The graph

Fix a prime power q and an integer $s \geq 2$. Put

$$R_s = \mathbb{F}_q[t]/(t^s).$$

Inside $SL_3(R_s)$ define

$$U_0 = \{I + atE_{31} : a \in \mathbb{F}_q\}, \qquad U_1 = \{I + aE_{12} : a \in \mathbb{F}_q\}, \qquad U_2 = \{I + aE_{23} : a \in \mathbb{F}_q\}.$$

Let

$$G_s = \langle U_0, U_1, U_2 \rangle \leq SL_3(R_s).$$

In this note we use the graph associated with U_0. The other two choices are cyclically analogous. Set

$$K = \langle U_1, U_2 \rangle.$$

Then

$$K = \left\{ \begin{pmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} : x, y, z \in \mathbb{F}_q \right\},$$

so K is the Heisenberg group of order q^3, whereas U_0 has order q. Also

$$U_0 \cap K = \{1\}.$$

Define a bipartite graph $Y_s = Y_s(q)$ as follows. The left vertices are right U_0-classes

$$L_s = G_s/U_0,$$

and the right vertices are right K-classes

$$R'_s = G_s/K.$$

For every $g \in G_s$ put an edge

$$gU_0 \; - - \; gK.$$

Equivalently,

$$gU_0 \sim hK \quad \Longleftrightarrow \quad gU_0 \cap hK \neq \varnothing.$$

This is just an incidence graph between two quotients of the same finite group. No topology or building language is needed.

Lemma 1 (Degrees)**.** *The graph Y_s is (q, q^3)-biregular.*

Proof. A left vertex gU_0 is adjacent to the vertices

$$guK, \qquad u \in U_0.$$

Since $U_0 \cap K = \{1\}$, these q vertices are distinct. Thus the left degree is q.

Similarly a right vertex gK is adjacent to the vertices

$$gkU_0, \qquad k \in K,$$

and these are all distinct, again because $U_0 \cap K = \{1\}$. Thus the right degree is $|K| = q^3$. □

For fixed q, the graph size grows with s. Indeed, let

$$w = (I + E_{12})(I + E_{23})(I + tE_{31}) \in SL_3(\mathbb{F}_q[t]).$$

This element belongs to the group generated by the three displayed root groups over $\mathbb{F}_q[t]$. Its characteristic polynomial is

$$\chi_w(\lambda) = (\lambda - 1)^3 - t\lambda^2.$$

Hence w does not have finite order over $\mathbb{F}_q(t)$. Therefore its images modulo t^s have orders tending to infinity: otherwise, some bounded positive power would be congruent to the identity modulo t^s for infinitely many s, forcing that power to be the identity over $\mathbb{F}_q[t]$. Consequently $|G_s| \to \infty$ as $s \to \infty$.

Thus for every fixed q we get infinitely many graphs, all with the same degrees (q, q^3).

2 The spectral statement

Let A_s be the normalized bipartite adjacency operator from right functions to left functions:

$$(A_s f)(gU_0) = \frac{1}{q} \sum_{u \in U_0} f(guK).$$

Its largest singular value is 1, with constant singular vectors. Write $\sigma_2(Y_s)$ for the largest remaining singular value.

Theorem 1. *For every prime power q and every $s \geq 2$,*

$$\sigma_2(Y_s) \leq \frac{2}{\sqrt{q}}.$$

Consequently the unnormalized second singular value satisfies

$$\lambda_2(Y_s) \leq \sqrt{q \cdot q^3}\, \sigma_2(Y_s) \leq 2q^{3/2}.$$

If $\Delta = q^3$ is the larger degree, then

$$\lambda_2(Y_s) \leq 2\sqrt{\Delta}.$$

The proof is short. We use one elementary local computation and one elementary "swap-walk" estimate for tripartite incidence structures.

3 The local computation: the affine plane

The relevant local graph is the following Heisenberg incidence graph. Let

$$H = \mathbb{F}_q^3, \qquad (x, y, z)(x', y', z') = (x + x', y + y', z + z' + xy').$$

Let

$$X = \{(x, 0, 0) : x \in \mathbb{F}_q\}, \qquad Y = \{(0, y, 0) : y \in \mathbb{F}_q\}.$$

Consider the bipartite graph $\mathcal{L}_q = CC(H; X, Y)$ whose two sides are right cosets of X and right cosets of Y, with incidence given by nonempty intersection.

Right X-cosets are parametrized by pairs $(y, z) \in \mathbb{F}_q^2$. Right Y-cosets are parametrized by pairs $(x, w) \in \mathbb{F}_q^2$, where the coset represented by $(x, 0, w)$ consists of

$$(x, y, w + xy), \qquad y \in \mathbb{F}_q.$$

Thus the incidence relation is

$$(y, z) \sim (x, w) \quad \Longleftrightarrow \quad z = w + xy.$$

Lemma 2 (Affine-plane spectrum)**.** *The normalized second singular value of $\mathcal{L}_q$ is exactly $q^{-1/2}$.*

Proof. The normalized adjacency operator from the (x, w)-side to the (y, z)-side is

$$(Af)(y, z) = \frac{1}{q} \sum_{x \in \mathbb{F}_q} f(x, z - xy).$$

Fix a nontrivial additive character $\psi : \mathbb{F}_q \to \mathbb{C}^\times$. For $\alpha, \beta \in \mathbb{F}_q$, put

$$f_{\alpha,\beta}(x, w) = \psi(\alpha x + \beta w).$$

Then

$$Af_{\alpha,\beta}(y, z) = \psi(\beta z) \cdot \frac{1}{q} \sum_{x \in \mathbb{F}_q} \psi((\alpha - \beta y)x).$$

If $(\alpha, \beta) = (0, 0)$, this is the constant singular vector with singular value 1. If $\beta = 0$ and $\alpha \neq 0$, the output is identically 0. If $\beta \neq 0$, the output is supported on the line $y = \alpha/\beta$ and has normalized L^2 norm exactly $q^{-1/2}$ times the normalized L^2 norm of the input. These characters form an orthonormal basis on the (x, w)-side, so the largest nontrivial singular value is $q^{-1/2}$. □

4 Why this local graph controls Y_s

Besides

$$K_0 := K = \langle U_1, U_2 \rangle,$$

define

$$K_1 := \langle U_0, U_2 \rangle, \qquad K_2 := \langle U_0, U_1 \rangle.$$

A direct multiplication calculation gives

$$K_0 \cap K_1 = U_2, \qquad K_0 \cap K_2 = U_1, \qquad K_1 \cap K_2 = U_0,$$

and

$$K_0 \cap K_1 \cap K_2 = \{1\}.$$

Think of the three quotient sets

$$G_s/K_0, \qquad G_s/K_1, \qquad G_s/K_2$$

as three colors of vertices. A group element $g \in G_s$ determines a triangle

$$(gK_0, gK_1, gK_2).$$

Since the triple intersection is trivial, these triangles are in bijection with G_s.

The edge of colors 1 and 2 determined by g is the class

$$g(K_1 \cap K_2) = gU_0.$$

Therefore the graph $Y_s = CC(G_s; U_0, K_0)$ is precisely the incidence graph between color-0 vertices G_s/K_0 and opposite edges G_s/U_0.

Now look locally around a color-0 vertex, say K_0. The adjacent color-1 vertices are the cosets of $K_0 \cap K_1 = U_2$ inside K_0, and the adjacent color-2 vertices are the cosets of $K_0 \cap K_2 = U_1$ inside K_0. Thus the link is

$$CC(K_0; U_2, U_1),$$

which is exactly the Heisenberg graph $\mathcal{L}_q$ from the previous section. The same calculation applies at every vertex and every color. Hence all local links have normalized second singular value $q^{-1/2}$.

We finish with the standard two-dimensional swap-walk estimate. It is included here in the only form needed.

Proposition 1 (Colored vertex-edge incidence estimate)**.** *Let X be a finite tripartite triangle system with the following regularity: triangles have one vertex of each color, every edge lies in the same number of triangles, and the uniform triangle measure induces the normalized random walks. Assume that every vertex link has normalized second singular value at most λ. Then, for each color i, the normalized incidence operator between color-i vertices and edges of the other two colors has nontrivial singular value at most 2λ.*

Proof idea. The square of this incidence operator is the two-step walk that starts at a color-i vertex, chooses a random opposite edge in its link, and then chooses a random color-i vertex completing that edge to a triangle. This is the "colored swap walk" $S_{i,\{j,k\}}$.

The usual proof is a one-line Hilbert-space estimate after conditioning on the middle vertex in the swap. On each conditioned link, the nonconstant part of the walk contracts by at most λ. The only remaining pieces are the two one-color shadows, one through color j and one through color k; applying the same link estimate to these two shadows gives two contributions, each at most λ. Hence the global nonconstant contraction is at most 2λ. Equivalently, for every mean-zero function f on opposite edges and every mean-zero function h on color-i vertices,

$$|\langle Af, h\rangle| \le 2\lambda \, \|f\|_2 \, \|h\|_2 \,,$$

where A is the normalized incidence operator. This is exactly the desired singular-value bound. □

Applying the proposition with

$$\lambda = q^{-1/2}$$

gives

$$\sigma_2(Y_s) \le 2q^{-1/2}.$$

This proves the theorem.

5 Parameter summary

For every prime power q and every $s \geq 2$, the graph $Y_s(q)$ satisfies

$$\text{left degree} = q, \qquad \text{right degree} = q^3,$$

$$|V(Y_s)| = |G_s|/q + |G_s|/q^3,$$

$$\sigma_2(Y_s) \leq 2q^{-1/2},$$

and

$$\lambda_2(Y_s) \leq 2q^{3/2} = 2\sqrt{q^3}.$$

For fixed q, the parameter s gives infinitely many graph sizes tending to infinity while the degrees and spectral bound remain unchanged.

Equivalently, for every $\Delta = q^3$ with q a prime power, there is an explicit infinite family of $(\Delta^{1/3}, \Delta)$-biregular graphs with unnormalized second singular value at most $2\sqrt{\Delta}$.